

Team Worksheet — Lab 18

Multiple Regression and Model Diagnostics (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-18-multiple-regression.xlsx`.

3. What we produced

Deliverable: A validated multiple regression model with full residual and multicollinearity diagnostics.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap simple linear regression: the fitted line, slope interpretation, R-squared and the correlation-is-not-causation trap.

Step 2. Extend to multiple regression: $Y = b_0 + b_1X_1 + b_2X_2 + \dots + e$, where each coefficient is the effect of that X holding all other Xs constant.

Step 3. Understand why that 'holding constant' clause matters — it is what makes multiple regression fundamentally different from running several simple regressions.

Step 4. Assemble your capstone dataset with the Y and at least four candidate Xs, with enough rows (a common rule of thumb is 10-15 observations per X).

Step 5. Fit the full model and record the coefficients, their p-values, R-squared and adjusted R-squared.

Step 6. Use ADJUSTED R-squared, not R-squared, to compare models. Plain R-squared always rises when you add an X, even a random one.

Step 7. Check multicollinearity with Variance Inflation Factors. $VIF > 5$ is a concern, $VIF > 10$ is serious — correlated Xs make coefficients unstable and signs nonsensical.

Step 8. Resolve multicollinearity by dropping or combining redundant Xs, then refit.

Step 9. Run the residual diagnostics in full: residuals versus fitted values (should show no pattern), normal probability plot of residuals, and residuals versus order (independence).

Step 10. Interpret residual patterns: a funnel shape indicates non-constant variance, curvature indicates a missing quadratic term, and drift over order indicates autocorrelation.

Step 11. Reduce to the final model using backward elimination — drop the least significant X, refit, and repeat until all remaining Xs are significant.

Step 12. Guard against overfitting: hold back a validation subset, or use cross-validation, and confirm the model predicts data it has not seen.

Step 13. Translate the final model into business language: for each significant X, state what a one-unit change does to Y in customer or dollar terms.

7. Self-check

Your final model has all VIFs under 5, residual plots show no pattern, adjusted R-squared is reported, the model has been validated on held-out data, and each coefficient is stated in business terms.

Complete? ■ Yes ■ Not yet

